SPMM Smart Revise

Perinatal Psychiatry

Paper B **Syllabic content 7.2**

1. Mental illness and pregnancy

Pregnancy is associated with risk of spontaneous major malformation (2-3% of all pregnancies) and drugs account for only 5 of every 100 malformations that occur.

Pregnancy is not a protective factor for psychosis and depression. In fact, pregnancy is associated with increased risk of suicide and mental health problems. This may be due to hormonal, social, personal changes and emotional stress. Treatment of mental health problems should be a priority but should be based on an individual risk-benefit assessment.

- A significant increase in new psychiatric episodes in first 3 months of the postpartum period. 80% are mood disorder (mainly depression)
- Depression during pregnancy – 7 to 15% risk (around 10%). This may be higher in developing nations. The rate of depression in a woman outside the perinatal period is only 7%. The relapse rate of depression in those who had a past history is around 50% when they are pregnant. The high prevalence of antenatal depression may be partly due to relapse/recurrences and partly due to freshly increased the incidence of depression. (O'Keane & Marsh 2007)
- Postpartum depression: 10% risk, begins before delivery, higher risk in patients with previous depressive illness (irrespective of being postpartum or not) & risk highest with bipolar illness.
- The risk of postpartum psychosis is 0.1-0.25% in general population, 50% in bipolar disorder and 50-90% in a patient with a history of postpartum psychosis.
- Puerperal psychosis can take the form of mania, schizophreniform / acute polymorphic psychosis, or with confusion and disorientation.
- The incidence of puerperal psychosis is about one per 1000 births.
- It is strongly linked with the bipolar disorder.
- Childbirth, together with abortion and menstruation are triggers of bipolar episodes
- The recurrence rate is about one in four pregnancies
- Disorders of the mother-infant relationship are prominent in 10–25% of women
- There is a 20-fold increase in relative risk of psychosis in one's lifetime during the first postpartum month. Women with the schizoaffective disorder are also susceptible.
- The risk of relapse of bipolar illness is increased up to eight-fold in the first month postpartum. Most relapses are depressive.

Risks of untreated psychiatric illness:

For the mother: Increased risk of suicide, alcohol & substance misuse, poor compliance with perinatal appointments, unhealthy lifestyle (poor diet, Lack of exercise, increased smoking), poor judgement, impulsive acts & impaired self-care.

For the foetus: Low birth weight & small head circumference (due to antenatal depression & anxiety), preterm birth (due to antenatal depression).If untreated depression continues in the postnatal period, it leads to attachment, behavioural & cognition difficulties. Substance misuse in pregnancy leads to increased intrauterine deaths, congenital, cardiovascular & musculoskeletal anomalies, & foetal alcohol syndrome. The harm due to postpartum psychosis can range from neglect to infanticide.

2. Pharmacological treatment in pregnancy

A. General principles of psychiatric treatment in pregnancy

Risks of pharmacological treatments in pregnancy include - Neonatal toxicity, prematurity and stillbirth, and morphological and behavioural teratology. Major malformations occur in the first trimester and neonatal toxicity in the 3rd trimester. Teratogenic effects are both dose and time dependent, with organs at the greatest risk during their period of fastest development. Week 6 to week 10 is the most vulnerable period.

- Treatment must be based on individual risk-benefit analysis
- Monotherapy, lowest dose, regular psychiatric and obstetric review & review of medications are recommended
- Involve midwives, obstetricians and health visitors. Refer to specialist perinatal services if necessary.
- If possible, avoid all drugs in the 1st trimester.
- Ensure adequate foetal screening during pregnancy
- Adjust doses as pregnancy advances; blood volume expands by 30% in the 3rd trimester. Hence, dose increase is frequently required in the 3rd trimester.
- Observe for neonatal withdrawal symptoms after birth

✶ **Treatment of schizophrenia in pregnancy:**

- Risk associated with untreated schizophrenia is high
- Consensus- use antipsychotic at every stage of pregnancy. Patients with a history of psychosis, who are maintained on antipsychotic, particularly if they have frequent relapses, are best maintained on antipsychotic during and after pregnancy.
- Most experience is with chlorpromazine, trifluoperazine, haloperidol, olanzapine and clozapine.
- Olanzapine is widely used by perinatal services in the UK.

✶ **Treatment of depression in pregnancy:**

- Explore the possibility of delaying the treatment until second or third trimesters
- Alternative treatment possibilities e.g. CBT should be explored
- Patient with a high risk of relapse and on an antidepressant should be maintained on an antidepressant during and after pregnancy.
- If a patient develops a moderate or severe depressive illness during pregnancy, they should be treated with antidepressant medication if psychological management has failed or is not available
- Nortriptyline, amitriptyline, impramine and fluoxetine are recommended antidepressants. Avoid paroxetine

✶ **Treatment of bipolar affective disorder in pregnancy:**

- Most of the patients who become pregnant while on medication and patient with severe illness & with high risk of relapse should be maintained on medication
- "Maintenance strategies should involve dosage reduction and regular review of side effects"(Kohen, 2004)
- "Discontinuation of mood stabilisers in pregnancy should take place only when absolutely necessary and be followed by frequent monitoring"(Kohen, 2004)
- "For women who have had a long period without relapse, the possibilities of withdrawing treatment before conception and for at least the first trimester should be considered". (Maudsley.2007)
- Avoid valproate and combination of mood stabilisers
- If Valproate or carbamazepine is prescribed; Prophylactic folic acid (5 mg daily, from at least a month before conception) should be used.
- Prophylactic Vitamin K should be given mother & neonate after delivery when Valproate or carbamazepine is used.

B. Specific effects of various drugs

✶ **Tricyclic Antidepressant**: They do not cause any significant malformation in all three trimesters. But preferably use nortriptyline & desipramine as they have less hypotensive and anticholinergic side effects. However, high dosages use (similar to fluoxetine) in the third trimester can lead reversible withdrawal symptoms (irritability, eating & sleeping difficulties and convulsions)

✶ ***SSRI***: No increase in major malformation (exception- paroxetine); the neonatal withdrawal is reversible complications. However there is 13.3% increase in spontaneous abortion (also with mirtazapine & bupropion), decreased gestational age (mean 1 week) and low birth weight (mean 175 gms).

- Paroxetine, particularly high dose first-trimester exposure, is clearly linked to cardiac malformation – VSD and ASD. Third-trimester use can give rise to neonatal complication due to abrupt withdrawals.
- Fluoxetine has the most evidence and seems safer than paroxetine.
- Least placental exposure is with sertraline.
- Exposure to SSRI, when taken in late pregnancy, may increase risk for persistent pulmonary hypertension of the newborn although the absolute risk is small
- Note: risk of neonatal withdrawal symptoms is high with paroxetine and venlafaxine due to short half-life

✶ ***MAOIs***: Evidence safety of MAOIs is very limited. Therefore switching to a safer antidepressant is advised. They should be avoided due to the risk of hypertensive crisis and suspected increase risk of congenital malformation.

✶ ***Other antidepressants***: Limited data on moclobemide, venlafaxine, reboxetine, Bupropion and mirtazapine suggests the absence of teratogenicity.

- ✶ ***Lithium:*** 1 in 10 chance of having a malformation if lithium is continued through the first trimester. The UK National Teratology Information Service have concluded that lithium increases the risk of all types of malformation of approximately three-fold and with a weighting towards cardiac malformations of around eight-fold (Williams & Oke, 2000)
    - Ebstein's anomaly-Relative risk compared to general population-10-20times higher, but the absolute risk is low at 1:1000. The (absolute spontaneous risk of Ebstein's is 1 in 20,000. Cohen et al., *JAMA* 1994;271:146-150) (9.5 is the closest answer to the question that appeared in March 2008)
    - Maximum risk is at 2-6 weeks after conception when many pregnancies are still undiscovered.
    - Foetal toxicity-Hypotonia, lethargy, poor reflexes, respiratory difficulties & Cardiac arrhythmias. (Note- these are reversible and do not cause later complication)
    - Important- Risk to mother and child of lithium withdrawal might have been underestimated and risk to the foetus of lithium exposure might have been overestimated.
    - The risk of relapse is up to 70% within 6 months, faster the discontinuation- higher the risk of relapse.
    - The risk of relapse on discontinuation during pregnancy is same for pregnant and non-pregnant women. But among women with bipolar disorder who elect to discontinue lithium therapy in the puerperium, the estimated risk of relapse is threefold higher than for nonpregnant, nonpuerperal women.
- ✶ ***Carbamazepine:*** 0.5 -1% risk of spina bifida, craniofacial anomalies, growth retardation and decreased the average head circumference. In a series of carbamazepine exposed pregnancies, craniofacial defects (11 percent), fingernail hypoplasia (26 percent), and developmental delay (20 percent) were recorded. There is some similarity between the teratogenic effects of carbamazepine and foetal hydantoin syndrome (phenytoin exposure) probably related to the common arene oxide pathway through which both drugs are metabolised. Hence the epoxide intermediate rather than carbamazepine itself may be the principal teratogenic agent.
- ✶ ***Sodium Valproate:*** Most teratogenic. The risk for any birth defect quoted 7.2% in Maudsley & NICE. UK epilepsy & pregnancy registry gives comparable value; North American registry quotes 10% risk. The risk is dose related and mostly seen in 17 to 30 days post conception; risk increases with family history of neural defects
    - Causes foetal distress, growth retardation, hepatotoxicity and congenital anomalies.
    - Congenital anomalies- risk of neural tube defect (1-2 %), risk of spina bifida (10 fold increase),digital and limb defects, heart defects (VSD, Pulmonary stenosis etc. 4 fold increase), urogenital malformations, low birth weight and psychomotor slowness
    - Causes neurological dysfunction and hyperexcitability, the severity of dysfunction correlates with valproate's concentration in the infant
    - An increasingly recognised risk in exposed children is mental retardation – low IQ due to Valproate exposure seems to be the most common result of the mother being treated with Valproate when pregnant. Among children whose mothers took valproate monotherapy

during pregnancy, 42% had a verbal IQ score of less than 80 (22%, <70 Adab et al., 2004). Interestingly in epilepsy literature, it is also recorded that having 5 or more seizures during pregnancy can significantly affect the verbal IQ of the newborn. So the confounding effect of seizures is not known. They also reported that 30% of children exposed to valproate needed special educational support in school, compared with 3-6% of those exposed to monotherapy with other antiepileptic drugs (Breen & Davenport, 2006)

- ***Lamotrigine*** may be associated with cleft palate. Lamotrigine monotherapy is associated with a 3.2% frequency of malformations.
- ***Conventional antipsychotics***: The exact risk of significant harm caused by antipsychotics is unknown.It is generally agreed that conventional antipsychotic are associated with low teratogenic risks and are comparatively safe to use in pregnancy. The most notable study was California child health development project which studied 19,000 births. But low-potency conventional antipsychotics are associated with the transient perinatal syndrome, floppy infants & withdrawal symptoms (Irritability, hypertonicity, hypotonicity & underdeveloped reflexes). Low potency used during the first trimester has been associated with a small but statistically significant increased relative risk of congenital malformation (2-2.4%). First-trimester haloperidol exposure has been associated with limb deformities however other studies did not conform this association (Yaeger et al., 2006)
- ***Atypical antipsychotics:*** Case reports suggest that clozapine, olanzapine, risperidone and quetiapine, in pregnancy, do not have serious effects on the newborn. There are concerns about the use of clozapine due to lack of foetal white cell measurement, lack of understanding of the risk of agranulocytosis and hypotension. Still birth & neonatal seizure have been reported with clozapine. Gestational diabetes can occur with clozapine & olanzapine. Note- Data on aripiprazole is limited. No case reports on use of other SGAs in pregnancy
- **Anticholinergic drugs:** Teratogenicity has been reported with anticholinergic medications (in a combination with antipsychotic) use in pregnancy. They should be avoided and if needed, the lowest possible dose should be used.
- **Benzodiazepines**: First trimester-0.6% risk of oral cleft & CNS and urinary tract malformation. Other adverse effects-Neonatal toxicity (withdrawal symptoms), respiratory depression, muscular hypotonia (floppy baby syndrome). Benzodiazepines are best avoided in pregnancy. Zopiclone does not show any animal teratogenicity. Human data are lacking. Note: - Promethazine is widely used as sedative in pregnancy but data is limited

Lithium treatment programme in pregnant mothers (Adapted from Kohen,2004)
•In women on maintenance treatment, serum lithium level should be monitored every 4 weeks through out the pregnancy •Lithium dosage should be adjusted to match the lower end of the therapeutic range •Lithium should not be discontinued abruptly; prior to delivery the dosage should be gradually tapered to 60-70% of the original maintenance level •Lower doses and frequent blood monitoring should be norm in pregnant women starting lithium in the first trimester of pregnancy •Lithium commenced in second and third trimester of pregnancy or perinatal period can help reduce the risk of puerperal psychosis •No decision is risk free •guidelines should vary with the severity of the illness •Should undergo level 2 ultrasound and echocardiography of the foetus at 6 & 18 week's gestation to screen Ebstein's anomaly.(Maudsley.2007) •Increased dose of lithium is required during 3rd trimester as total body water increases, but the requirement returns abruptly to pre-pregnancy level immediately after delivery.

ECT in pregnancy: Relatively few side effects and fewer risks than untreated mood episodes or pharmacotherapy. Occasional reports of congenital malformations; but does not implicate ECT as a causal factor. Complications of ECT during pregnancy are uncommon and transient

- Effect of anaesthetic agents: Barbiturates and atropine can reduce beat-to-beat variability in the foetal heart rate, and atropine can cause foetal tachycardia.
- Uterine smooth muscle does not routinely contract during a seizure so premature labour is almost unheard of.
- Seizure threshold is decreased by oestrogen and increased by progesterone.
- During pregnancy, prolonged gastric emptying time increases the risk of gastric regurgitation and aspiration pneumonitis (Yonkers et al. 2004)

3. Pharmacological treatment and breastfeeding

- ✶ Systematic research and controlled studies on the effect of psychotropics on nursing mother and baby are lacking and recommendation are based on small case series and single case reports. Long term adverse effects of psychotropic medication is not available.
- ✶ All psychotropic are excreted in breast milk. The risk-benefit ratio should be considered on individual basis. Treatment of maternal illness should take priority wherever needed. Monotherapy with lowest possible dose should be used. Medication should be taken after breast-feeding and should be avoided during peak drug level in milk

A. General pharmacokinetics of psychotropics in breast milk

Factors playing a role in development of adverse effect in breast fed infants are the prescribed dose; level of drug the mother's blood plasma; the level in the breast milk; and the level in the infant's serum. The factors determining the amount of drug excreted in the milk include a medication's diffusion capacity across the membranes, its molecular weight and lipid solubility.

- "The arbitrary concentration in the infant's plasma of 10% of the established therapeutic maternal dose is used as the upper threshold where the risks of a particular drug side effects are low and treatment is accepted as safe"
- A breast milk/ mother's plasma ratio greater than 1 suggests a high likelihood the infant will be exposed to the drugs
- Pre-term immature infants should not be exposed to psychotropic medication as they are more sensitive and may have an immature liver function.
- Infants older than 10 weeks are at lower risk of adverse effects if there is no evidence of accumulation in the infant.
- The infant's cardiac, renal, and hepatic function should be checked before breast-feeding.
- If breastfed, infants progress, milestones and adverse effect (drowsiness, hypotonia, rigidity, tremor and withdrawal symptoms) should be monitored.
- Colostrum will have greater concentration of protein-bound drugs and hind milk will have greater concentration of lipid soluble drugs compared to foremilk
- **Factors to be considered in risk/benefit analysis:** Severity and frequency of mental illness, benefits of breastfeeding, impact of untreated maternal illness on infant and mother, level of family support, compliance with treatment, patient's and family's ability recognise early warning signs, physical health and maturity of the infant, support from statutory and voluntary organising
- **Recommendation** (Adapted from Maudsley guidelines, 2007)

Antidepressant- paroxetine or sertraline
Antipsychotic -Sulpride or olanzapine
Mood stabiliser- avoid if possible, Valproate if essential
Sedatives- lorazepam for anxiety; zolpidem for sleep

B. Specific effects of various drugs

Antidepressants: These are secreted in breast milk in very small quantities. Infant serum levels are low.

- SSRI (Fluoxetine, sertraline, paroxetine and citalopram) and tricyclic antidepressant (except Doxepin) are probably safe.
- Preferred Tricyclic antidepressants-Amitriptyline and imipramine.
- Amitriptyline, imipramine, clomipramine and nortriptyline did not show any adverse effects in infants. Maternal plasma and breast milk concentration of tricyclic antidepressant is same. N-desmethyldoxepine, longer acting metabolite of doxepin, may accumulate in infants and cause severe drowsiness and respiratory depression.Plasma levels of clomipramine in breast fed infant were high following delivery.
- Data is limited for trazodone and should be used with caution.
- Fluoxetine and its active metabolite norfluoxetine were detected in plasma and breast milk but not in infant's plasma. Single case reports have indicated adverse effects such irritability, cyanosis, somnolence and unresponsiveness with higher doses. It does not effect the development; does not cause cognitive dysfunction or neurological abnormality.
- Sertraline is the first line of treatment in the USA. The samples for sertraline studies are large.
- Paroxetine has a lower milk/plasma ratio than fluoxetine and sertraline.
- No studies of MAOI's or bupropion use in breastfeeding are available. MAOI's should be stopped in mothers planning to breast-feed.

Median Time to maximum concentration in the milk after maternal ingestion
Moclobemide- 3 hours
Olanzapine- 5 hours
Sertraline-7-10 hours, Peak level have not been reported for paroxetine or fluoxetine

Conventional Antipsychotics: High-potency antipsychotic are recommended during breast-feeding due less sedation and less autonomic effect. Haloperidol, chlorpromazine, or perphenazine have been studied and they are safe during breastfeeding. Delayed development has been reported in 3 infants who were

exposed to a combination of haloperidol and chlorpromazine (Maudsley guidelines,2007).

Atypical antipsychotics: Limited data suggests sulpride; olanzapine and risperidone are safe in breastfeeding. However, the decision to continue needs to be done on individual risk/benefits analysis. There are no published data on Aripiprazole, Amisulpride, Sertindole and Ziprasidone. Clozapine is contraindicated during breastfeeding as it accumulates in breast milk and foetal serum. A Higher concentration of clozapine is due to a higher concentration of albumin in the foetal blood. Reported adverse effects - Agranulocytosis (but recovered spontaneously after discontinuation), decreased sucking reflex, drowsiness seizure, irritability and cardiovascular instability.

Lithium*:* Lithium is contraindicated during breastfeeding (Kohen,2005).However, “the use of lithium during breastfeeding varies between absolute contraindication to mother's informed choice” (Maudsley Guidelines, 2007).Lithium serum concentration and CBC should be monitored for infants exposed to lithium.

- 40- 50% of maternal serum level is excreted in breast milk.
- Infant serum level can rise up to 200% of maternal serum concentration (range 5%-200%)
- Serum level of lithium can be raised due to diminished renal clearance in neonates
- Toxicity has been reported -cyanosis, lethargy, Hypotonia, heart murmur (these symptoms resolved after stopping the breastfeeding)

Sodium Valproate*:* Infant serum levels range from being undetectable to 40%. Adverse effects reported-Thrombocytopenia and anaemia (resolved after stopping breastfeeding)

Carbamazepine*:* Infant serum range- 5-65%. Adverse effects reported-Cholestatic hepatitis, transient hepatitic dysfunction (Hyperbilirubinaemia, High concentration of gamma-glutamyl-transferase), seizure-like activity, irritability, high-pitched crying, hyperexcitability and poor feeding

Lamotrigine*:* Infant serum range- 30% of maternal concentration. No adverse effects noted

Benzodiazepines should not be started during breastfeeding and should be encouraged to stop before becoming pregnant. A long-acting benzodiazepine can produce lethargy, poor suckling and weight loss. Persistent apnoea has been reported with Clonazepam. Infant serum range- Diazepam: undetectable-15%, Lorazepam, clonazepam and temazepam: small amount. A low dose of short-acting benzodiazepine (Temazepam, oxazepam) are mostly safe. Diazepam and alprazolam should be avoided.

Buspirone, zaleplon and zopiclone are excreted in breast milk and should be avoided. Zopiclone: Infant serum range - up to 50% of the maternal plasma level. Zolpidem is safe during breastfeeding

Notes prepared using excerpts from

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